

First-Principles Generation Mechanism of Spacetime, Force and Mass Based on Zero Global Angular Momentum Axiom

Abstract

Traditional physics treats spacetime, fundamental force, and inertial mass as independent primitive quantities of the universe, lacking unified first-principles derivation. Based on the core axiom of ANG-TOE (global total angular momentum equals zero), this paper constructs a complete generation logic of macroscopic physical quantities. The universe's noumenon is a six-dimensional angular momentum topological manifold without innate spacetime, mass, or force. All classical physical observables are projective emergent phenomena of local angular momentum network. Local projection of angular momentum generates three-dimensional vector force fields and angular momentum exchange rates; closed-loop topology of three-dimensional force fields produces inertial mass; the decoupling mapping of three-dimensional exchange rates constructs a six-dimensional time-frequency-phase perception space; the intrinsic geometric coupling coefficient of space and time dimensions yields the speed of light. This work completely unifies the origin of spacetime, force, mass and light speed under a single axiom, realizing the closed-loop physical derivation of core macroscopic ontology.

Keywords: ANG-TOE; zero angular momentum axiom; projective emergence; spacetime origin; mass generation; topological field

1. Introduction

Classical mechanics, relativity, and quantum mechanics all take spacetime and mass as predefined fundamental background quantities, failing to explain their physical origin. Modern field theory constructs interactions based on existing spacetime backgrounds but cannot derive spacetime itself from more primitive geometric rules.

The ANG-TOE theoretical system breaks through the traditional physical paradigm: the universe has no innate spacetime, no primitive mass, and no independent fundamental force. The only primordial rule is the conservation of zero global angular momentum. All macroscopic physical entities are low-dimensional projection emergent effects of high-dimensional angular momentum network topology.

This paper strictly deduces the full generation chain of **force** → **mass** → **time-frequency-phase space** → **spacetime coupling constant (speed of light)** from the unique first principle, establishing a self-consistent unified generation theory of core cosmic physical quantities.

2. Primitive Axiom and Noumenon Definition

2.1 Core Axiom

The total angular momentum of the closed cosmic system is permanently zero:

2.2 Cosmic Noumenon

The cosmic primitive noumenon is a six-dimensional pure angular momentum topological manifold $\mathcal{M}_6^{\text{phys}}$, which only contains vortex-antivortex complementary link structures and angular momentum flux transfer relations. There are no pre-existing concepts of space, time, mass, energy, or force in the noumenon domain.

All observable physical quantities are generated by the projection operator $\mathcal{P}$ · mapping the high-dimensional topological structure to the human perception dimension.

Let Ω be any finite local subnet of the manifold, and τ be the primitive topological order parameter of the noumenon manifold, which only describes the sequence of topological evolution and does not belong to observable time.

3. Generation of Three-Dimensional Fundamental Force Field

3.1 Local Projection of Angular Momentum

After the local angular momentum tensor J_Ω of the six-dimensional noumenon is projected to the three-dimensional perception domain, two core physical quantities are resolved:

where J_3 is the three-dimensional spatial angular momentum vector field after projection; $\frac{dJ_3}{d\tau}$ is the three-dimensional orthogonal angular momentum flux exchange rate vector, representing the flux transfer rate of the local network in three orthogonal spatial directions.

3.2 Definition and Generation of Fundamental Force

Macroscopic force is defined as the spatial gradient constraint of local angular momentum density:

V denotes the projected local three-dimensional volume element.

Physical Proposition: Force is not an independent primitive interaction. It is the spatial gradient projection reading of local angular momentum field variation. All fundamental forces in nature originate from the topological gradient difference of angular momentum network projection.

4. Topological Generation Mechanism of Inertial Mass

4.1 Closed-Loop Constraint of Local Force Field

For any bounded local domain, if the angular momentum link forms self-winding closed topology and cannot completely dissipate outward, the force field satisfies the non-zero closed surface integral condition:

4.2 Topological Definition of Mass

The inertial mass of an object is strictly equivalent to the topological locking strength of the three-dimensional force closed-loop structure:

4.3 Physical Essence of Mass

Mass is not an inherent attribute of matter, but a **topological invariant of force field closed-loop constraint**. The higher the degree of force field closure and the stronger the local topological binding, the greater the inertial mass measured externally.

When the closed-loop topology is broken, the force field constraint is released, mass defect occurs, and the stored angular momentum flux is released outward in the form of field propagation.

5. Mapping Construction of Time-Frequency-Phase Space

Time, frequency and phase do not exist in the primitive noumenon domain, and all time-frequency-phase degrees of freedom are completely derived from the decoupling transformation of three-dimensional angular momentum exchange rate.

5.1 Projection from Topological Order to Observation Time

The primitive topological sequence differential $d\tau$ has no uniform flow characteristics. After projective transformation:

The discrete topological sequence is transformed into a continuous and uniform one-dimensional observation time base, which completes the generation of macroscopic time dimension.

5.2 Decoupling Transformation of Three-Dimensional Exchange Rate

Chain differentiation is performed on the angular momentum exchange rate under the observation time benchmark:

The time-domain flux variation rate is orthogonally decomposed into frequency component and phase evolution component:

where $\omega = (\omega_x, \omega_y, \omega_z)$ is the three-dimensional orthogonal frequency base formed by angular momentum oscillation exchange; $\frac{d\phi}{dt}$ is the global topological phase evolution rate, which generates the independent phase dimension ϕ .

5.3 Completion of Six-Dimensional Perceptual Spacetime Manifold

Three-dimensional spatial base x, y, z and three-dimensional frequency base $\omega_x, \omega_y, \omega_z$ linearly span the complete perceptual phase space:

The phase variable is embedded in all lattice points of the phase space as a hidden topological attribute, forming the complete physical carrier of human macroscopic observation: **time-frequency-phase integrated spacetime manifold**.

6. Geometric Origin of Spacetime Coupling Coefficient (Speed of Light)

In the six-dimensional time-frequency-phase space, the spatial line element and time differential element satisfy the rigid geometric constraint of angular momentum perturbation propagation:

The first-principles definition of the speed of light is derived:

The speed of light is not an artificially set cosmic constant nor an independent physical quantity. It is the **innate geometric coupling coefficient between spatial dimension and time dimension in the projective spacetime manifold**. Its value is uniquely determined by the topological projection rule of the high-dimensional angular momentum network. Local spacetime distortion caused by topological supersaturation will lead to regional deviation of light speed.

7. Conclusion

Under the single zero global angular momentum axiom of ANG-TOE, this paper completes the rigorous first-principles derivation of the generation of spacetime, fundamental force, and mass:

1. The primitive universe is only a six-dimensional angular momentum topological network without any macroscopic physical background;
2. Local angular momentum projection generates three-dimensional force fields, and force field gradient is the essential origin of all fundamental interactions;
3. The closed-loop topological locking of three-dimensional force fields produces inertial mass, realizing the geometric origin of mass;
4. The decoupling mapping of angular momentum exchange rate constructs a complete six-dimensional time-frequency-phase spacetime system;

5. The proportional coupling relationship between space and time dimensions defines the geometric constant of light speed.

This theory eliminates all empirical assumptions of traditional physics on spacetime and mass, unifies the most core macroscopic physical ontology under a single geometric axiom, and establishes a complete and closed theoretical foundation for the unified theory of everything.

References

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